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Provably Label-Efficient Conformal Prediction

A Sequential Stopping Rule Cuts Calibration Labeling Cost by 41% Without Sacrificing Coverage

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Conformal prediction's finite-sample coverage guarantee is usually presented as free once a calibration set exists—but collecting those labels rarely is. This paper treats calibration labels as a costly, queryable resource and designs an online stopping rule that automatically balances labeling cost against prediction-set efficiency. The rule matches the best *fixed* calibration size chosen with hindsight, cuts total labeling cost by **41.4%** ± **2.3%**, and provably preserves exact coverage despite deciding, on the fly, when to stop labeling.

AT A GLANCE

Problem: Conformal calibration-set size is treated as a free given, but labels needed to build it are often expensive to collect.

Why it matters: Medical labeling, expert annotation, and expensive simulation make calibration cost the dominant cost of deploying conformal prediction in practice.

Method: An online, cost-aware stopping rule for sequential label queries that halts once further labels are provably not worth their cost.

Result: Matches the best fixed calibration size in hindsight; cuts total labeling cost by 41.4% ± 2.3% empirically.

Evidence: Benchmark classification and regression tasks comparing adaptive stopping against fixed calibration-set-size baselines, with coverage validated throughout.

The Big Picture

Split conformal prediction fixes a calibration set of size n up front; more calibration data generally tightens prediction sets, with diminishing returns, and every additional label carries a real acquisition cost. Conformal theory has always treated n as given, leaving practitioners to guess a calibration budget with no statistical guidance. This paper asks the practical question directly: given unlabeled examples arriving one at a time and the ability to query a label at a cost, when should calibration stop and prediction begin?

Why Existing Approaches Fall Short

Practitioners today either guess a calibration-set size in advance, risking wasted labeling budget or an under-calibrated, oversized prediction set, or they collect labels until budget runs out with no statistical link between that stopping point and the tradeoff they actually care about. Naively stopping "when the sets look tight enough" is an adaptive, data-dependent rule that would ordinarily void the exchangeability conformal prediction relies on—so ad hoc adaptive stopping has been avoided rather than solved.

Core Mechanism

The authors formalize a total-cost objective combining the cumulative cost of labels queried so far with the expected inefficiency (prediction-set size) of the conformal predictor that would result from stopping now. They design a sequential stopping rule that halts querying once the estimated marginal benefit of one more label—in expected set-size reduction—no longer exceeds its cost. The delicate technical contribution is showing that this data-dependent stopping decision can be constructed so the resulting calibration set still yields a conformal predictor with exact finite-sample coverage, regardless of the adaptive stopping.

Technical Formulation

With per-label cost c and a weight λ trading off cost against expected prediction-set size $\mathbb{E}[|C(X)|]$, define the total objective after m queried labels as

$$J(m) = m \cdot c + \lambda \cdot \mathbb{E}\big[|C_m(X)|\big],$$

where C_m is the conformal predictor calibrated on the first m labels. The stopping time is

$$\tau = \inf\{m : \widehat{\Delta}(m) \leq 0\},$$

where $\widehat{\Delta}(m)$ estimates the marginal value of querying label $m+1$. Under mild regularity conditions, $\mathbb{E}[J(\tau)]$ matches $\min_m J(m)$—the best fixed calibration size chosen with hindsight—up to lower-order terms, while $P(Y \in C_\tau(X)) \geq 1-\alpha$ holds exactly despite τ being a data-dependent stopping time.

Learning or Inference Procedure

1. Observe an unlabeled example; decide whether to query its label based on the current estimate of marginal value $\widehat{\Delta}(m)$.

2. If $\widehat{\Delta}(m) > 0$, query the label, add it to the calibration set, and increment m.
3. If $\widehat{\Delta}(m) \leq 0$, stop querying at $\tau = m$.
4. Calibrate the conformal predictor on the τ collected labels and deploy it with the standard coverage guarantee.

What the Guarantee Says

Despite τ being chosen adaptively from the data observed so far, the resulting conformal predictor satisfies $P(Y \in C_\tau(X)) \geq 1 - \alpha$ exactly, in finite samples. Separately, the expected total cost at the stopping time matches the best achievable cost under a fixed calibration budget selected with hindsight knowledge of the data.

Experimental Findings

Across benchmark classification and regression tasks, the proposed stopping rule reduces total labeling cost by **41.4%** $\pm$ **2.3%** relative to fixed calibration-set-size baselines chosen without this guidance, while maintaining nominal coverage throughout the evaluated tasks.

Ablations and Interpretation

Cost sensitivity: As per-label cost c increases relative to the set-size weight λ, the stopping rule halts earlier, trading a somewhat larger prediction set for lower total cost—exactly the intended behavior.

Robustness to misestimation: Errors in estimating $\widehat{\Delta}(m)$ affect the total-cost optimality of the stopping point but never the coverage guarantee, which is preserved by construction regardless of how τ is chosen.

Practical takeaway: Calibration-set size, long treated as a free design choice in conformal prediction, can instead be solved for directly as a cost-aware stopping problem.

Reference

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